

Portable Sink Design

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Element A

A1: Whitemarsh Special Education program is in need of a portable sink that can act as a transitional tool to eventually build the students confidence to use the generic sink. The students often lack the ability to transition from one activity to the next. They require remediations to help them break down the tasks into more manageable tasks, that can eventually be pieced together to allow them to perform actions by themselves with assurance. The need for a portable sink stems from the desire to have students learn the mechanisms and steps in brushing their teeth, and washing their hands. Ultimately a successful designed portable sink would allow the teachers to break down the function of brushing teeth and washing hands in an environment that makes the students comfortable and most likely to retain information and mimic the teachers actions. The need for a portable sink also stems from the need for a tool that can break down the process of learning personal hygiene into a multistep process. Currently, the students are required to use the sink in the back of the classroom, which like said above, has caused difficulty for students, as this process can often be overwhelming to walk through.

A2: The stakeholders who are involved in the process and result of this project are the teachers as well as the current students within the program. They have and will continue to be the main stakeholders, who will impact the physical characteristics of the tool as well as the results of our testing and analysis of the product's performance. Their testing as well as the eventual permanent implementation of the product will all be conducted within the constraints of the classroom. The testing and use of the product also will be conducted during the constraints of public school hours, as well as any special events that may involve the use of learning tools outside of the general educational school day. The reason for the constraint of the school day is due to the fact that the tool will aid in visual learning within the classroom. Future projects may present the ability for a tool like this to be implemented within the out of classroom settings where students that need to break down these daily tasks can do that outside of the classroom. For now, the main focus is to design this tool within the needs of the classroom, and will continue to base all further decisions and plans to better the learning environment in the Whitemarsh Special Education Program.

A3: Creating a functional portable sink, can also be utilized in other environments. Within the past few years, mini-homes as well as vehicle-camping have become an intriguing lifestyle for people seeking adventure. Many people look for market items that imitate the full scale machines such as dishwashers. For this reason, a functional portable sink would likely be utilized in this industry, and would provide the functionality of a full scale sink. However, this product also could bring consequences to being advertised to this industry. Due to the need for durability in products within this exploration industry, because of exposure to extreme conditions, this could cause issues in the design of this product as it will be more focused on the classroom environment of the classroom rather than other external environments. Also, this problem can also be applicable to the general children's learning toys market. Many parents are responsible for advancing their infants in their early education. The focus of this project would likely have crossovers to these market stakeholders, and could possibly be successful in the children learning toys market. Overall, the creation of a portable sink can have many crossovers and benefits to other stakeholders and industries, so it may be important to consider these benefits within the design, creation, and evaluation of the product.

A4:

Primary Stakeholders:

- Whitemarsh Special Education teachers who enforce the learning activities and strategies within the classroom by demonstrations of how to use life-related tools
- The Whitemarsh Special Education students who learn and operate learning tools within the teachers supervisors
- Schools, including W.E.S., interested in purchase of portable sinks suitable for students with learning disabilities

Secondary Stakeholders:

- Parents of children within the program provide feedback and reflections on how their child is performing outside of the classroom, and any possible changes they notice in their children's habits/behaviors

Concerns:

- Due to the teacher's concern of portability, there is a need for a sink that can be held and carried from point to point.
- Due to the teachers concern of replication, there is a need for a sink that replicates a fully functional sink
- Due to the teachers concern of sensory overload for students, there is a need for simplistic functionality and appearance
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A5:

[The Outstanding Advantages of a Toddler Sink](#)

[Perceptual-motor coordination in persons with mild intellectual disability](#)

- A statistical experiment gathered 42 adults with Intellectual disabilities, and 48 age-matched typical adults.
- All participants performed each task (Box-and-Blocks, 25-Grooved-Pegboard, Stick Catching and overhead Bean Bag-Throw). All tasks were performed twice (one on each hand)

- Typical adults scored a significantly better score on one hand than the other, while the Adults with intellectual abilities showed no significant change in score on each hand during the tasks.
- Majority of the tasks yielded moderately-strong coefficients between the person, and their hand dexterity

On the relationship between motor performance and executive functioning in children with intellectual disabilities

- The Intellectual Disabled cohort had significantly scored lower on all assessments compared to the control group (Typical functioning children)
- However mild intellectual disabled children did not score significantly less on the object control skills test than borderline ID children.
- Children with intellectual disabilities often struggle more with tasks that require hand dexterity than children with no intellectual impairments.

Can Physical Exercise Promote the Development of Teenagers' Non-Cognitive Ability?-Evidence from China Education Panel Survey (2014-2015)

This article explains the importance motor development plays in physiological and sociological development. The second article shows that physical activity benefits learning abilities from children. From this, the first article hints that cognitive development, such as washing hands and brushing teeth may provide similar benefits.

The Direct Connection Between Learning Experience and Retention

- Connecting real-world applications and scenarios can enhance the learning experience and retention of knowledge
- Learners will be most likely to remember information when it relates to real world contexts
- Edtech (educational technology) is also on the rise and with the advent of this new technology, it will also create a more effective and engaging learning experience.
- Tasks that have been proven to improve long term retention on specific tasks, abilities, and information include repeated revisiting of material. Other tasks include a technique

called Interleaving. This includes the mixing of different subjects simultaneously, to better understand the material as a whole over time.

How Does Transitioning To A New Location Affect Children with Developmental Disabilities - 24 Hour Home Care

- Transition impacts children in many different ways (Not one child reacts exactly the same)
- Coping mechanisms play a major role in the response of the children to the pressure of transition
- Once children enter schooling, the responses to the many rapid changes can intensify
- Having special education programs allow for children to have access to handling transition in tangible ways, that ultimately condition them to respond appropriately to change
- Repeated enactments of the transitional task that needs to be learned often helps children ease the nerve of their uncertainty of the task asked of them
- Using mental, emotional, and physical tools helps children with these impairments to perform tasks over time by learning new work around to help them better understand and perform the tasks asked of them or required of them.
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How To Troubleshoot Low Water Pressure in Your House.

- The first step in addressing water pressure issues is to determine whether the problem is localized to specific fixtures, like a shower or sink, or if it affects the entire house. This distinction helps in identifying whether the issue is with individual components or the broader plumbing system.
 - Fixture-Specific Solutions:
- If low water pressure is isolated to a particular fixture, such as a shower, the article recommends checking for common culprits like a blocked shower head. Hard water deposits, visible as white clumps and cloudy spots, can be cleaned by soaking the showerhead in a vinegar-water solution overnight or by replacing it with a high-pressure alternative.
 - Sink Faucet Issues and Solutions:

- Clogged aerators and cartridges are highlighted as common causes of low water pressure in sink faucets. The article provides step-by-step instructions for cleaning the aerator using a toothbrush dipped in water and vinegar, as well as addressing clogged cartridges by disassembling and cleaning them.
 - Systemic Plumbing Evaluation:
 - If the low water pressure affects the entire house, the article suggests a comprehensive evaluation of the plumbing system. Possible causes include partially closed main shut-off valves or a malfunctioning pressure regulator. Homes over 20 years old with steel or iron pipes may require more extensive solutions like repiping with modern materials.
 - Hot Water-Specific Considerations:
- The article advises homeowners to differentiate between hot and cold water pressure issues. If the problem is specific to hot water, it recommends checking the shut-off valves near the hot water heater and ensuring they are fully open. Sediment buildup or blockages in the lines connecting to the water heater may require professional inspection and resolution.

[Food Safety Tip: Clean the kitchen sink.](#)

Study Findings on Sink Contamination:

- A USDA study mentioned in the article tested surfaces in kitchens where participants prepared a meal involving raw sausage, shell eggs, and a fruit salad with cut-up cantaloupe. Shockingly, the study revealed that 34% of sinks were contaminated after meal preparation, emphasizing the potential presence of harmful bacteria.

Cross-Contamination Risks:

- The article emphasizes the risk of cross-contamination from the sink to other items, including those consumed raw. Bacteria present in the sink or on hands can transfer to food items, highlighting the importance of maintaining a clean and sanitized sink environment.

Contamination of Cut-Up Cantaloupe:

- Notably, 26% of the samples of cut-up cantaloupe from the fruit salad were found to be contaminated after breakfast preparation. This underscores the need for thorough cleaning and sanitation practices, especially when dealing with fresh produce that is often consumed raw.

Cleaning and Sanitizing Recommendations:

- After using the sink for washing or preparing food, the article suggests a two-step cleaning and sanitizing process. First, wash the sink with warm, soapy water and wipe it clean with paper towels. Second, use a sanitizer (homemade or commercial) and allow the sink to air dry. This routine

helps eliminate bacteria and ensures a safer food preparation environment.

A6:

The proceeding of this project is of great importance to the targeted stakeholders of this project. The difficulties that the majority of children that face intellectual disorders encounter involve their hand dexterity, as well as how long it takes them to recall tasks asked of them. Statistical studies have shown that many intellectually disabled children learn differently and at a different pace than the normal child. Along with that, it's been determined that many children with intellectual disabilities often handle transition from one task to another with greater difficulty. Further, it's critical that we build a portable sink for the students of Whitemarsh Special Education Program, as their current stage in learning development requires them to learn how to wash their hands independently.

Moreover, having a sink that can mitigate the transition between a task to the sink across the room, can better help the students to stay focused and calm when transitioning from a previous task. Ultimately, with the information gathered above, we have chosen to proceed with this project, and believe the results of this project will enhance the learning environment for the students within the classroom.